

VM Cost Calculator

Calculation Methodology

Authoritative reference for how the calculator derives monthly recurring cost per virtual machine. Every formula in this document is transcribed from the shipped implementation in `app.js` so that the pricing output can be independently audited and reproduced by hand.

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Implementation reviewed	<code>app.js</code> — <code>costVm()</code> , <code>allCosts()</code> , <code>defaultPricing()</code> , <code>tbOf()</code> , <code>isWin()</code> , <code>r2()</code> , <code>usd()</code>
Live application	https://vmcost-vmw.pplx.app
Source repository	https://github.com/ryanwetmore492/vm-cost-calculator
Currency / period	USD, monthly recurring (MRC)
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1. Overview

The calculator is a pure client-side model of an MSP billing sheet for a VMware Cloud Director environment. It computes one monthly figure per VM as the sum of six independent charge lines, then aggregates those per-VM figures into grand totals, per-location totals and a SKU roll-up.

```
COST MODEL — costVm(vm), app.js
monthly_cost(vm) = compute
    + vmware_licensing      (global toggle)
    + storage
    + windows_spla          (OS contains "windows")
    + add_ons
    + zerto_dr              (per-VM toggle)
```

All rates are user-configurable. The values shipped in `defaultPricing()` are placeholders for demonstration only — they are not contract rates and must be replaced with the client's negotiated SKU pricing before any quote is issued. Every VM attribute (RAM, provisioned disk, OS string, ratio tier, storage tier, location, DR flag, DR GB, selected add-ons) is entered manually or imported from CSV.

Default rate catalogue (placeholders)

SKU	Charge line	Default rate	Unit / basis
2721	Enterprise Cloud Compute (4:1 processor ratio)	\$10.00	per GB RAM / month
CUSTOM	Enterprise Cloud Compute (2:1 processor ratio)	\$15.00	per GB RAM / month
2729	VMware Licensing	\$10.00	per GB RAM / month (enabled by default)
3819	Enterprise Cloud Storage — Standard Flash	\$333.00	per TB provisioned / month

SKU	Charge line	Default rate	Unit / basis
3815	Enterprise Cloud Storage — High Performance Flash	\$359.00	per TB provisioned / month
2589	Microsoft Windows Server Licensing (SPLA)	\$99.00	per Windows VM / month
(blank)	DR Storage — Zerto Replication	\$0.15	per DR GB / month
(blank)	Zerto Replication Fee	\$25.00	per protected VM / month

Global settings: GB→TB divisor = 1024 (default) or 1000; rounding mode = round each line item (default) or round totals only. DR SKU codes ship blank and are editable.

2. Charge lines

Notation: RAM_GB = configured memory in GB; Disk_GB = provisioned (allocated) disk in GB; divisor = the GB→TB setting; round() = the rounding mode function defined in §4.

2.1 Compute — SKU 2721 (4:1) and additional ratio tiers

Each VM is assigned exactly one processor-ratio tier. Compute is billed strictly on configured RAM; vCPU count is not an input to the model. The ratio tier only selects a per-GB rate — the ratio itself (4:1, 2:1, ...) is a label, not a multiplier.

```
compute = round( RAM_GB × ratio_tier_rate )

units:    GB × ($/GB/month) = $/month
applies:  always (every VM has a ratio tier)
```

Edge case: if the VM's ratioId no longer matches any tier in the catalogue (tier deleted), the rate resolves to 0 and compute is \$0.00 — the tier shows as “— none —” in the results table.

2.2 VMware licensing — SKU 2729

A second per-GB-RAM charge, governed by one global toggle (vmwareLic.enabled) that applies to all VMs at once. There is no per-VM override.

```
vmware = round( RAM_GB × vmware_lic_rate )    if vmwareLic.enabled
vmware = 0                                     otherwise

units:    GB × ($/GB/month) = $/month
applies:  all VMs, or none
```

2.3 Storage — SKU 3819 (Standard Flash) and 3815 (High Performance Flash)

Each VM is assigned one storage tier. A tier is priced either **per TB** (default) or **per GB**, and that choice determines the billed quantity:

```

tier priced per TB:   storage_qty = Disk_GB ÷ divisor
tier priced per GB:   storage_qty = Disk_GB           (divisor bypassed)

storage = round( storage_qty × storage_tier_rate )

units:   TB × ($/TB/month) or GB × ($/GB/month) = $/month
applies: always; billed on provisioned disk, not consumed disk

```

Edge cases: (a) a per-GB tier **bypasses the divisor entirely** — no GB→TB conversion is applied, so changing the divisor has no effect on those tiers; (b) tiers saved before the per-GB feature existed have no unit field and are normalised to per TB; (c) if the VM's storage tier is missing, quantity defaults to the TB basis and the rate resolves to 0, yielding \$0.00.

2.4 Windows Server SPLA — SKU 2589

A flat per-VM licence fee, applied automatically from the OS string. The test is a case-insensitive substring match — the regular expression `/windows/i` against the VM's OS field. Any OS text containing “windows” in any casing qualifies; there is no manual override and no per-core or per-socket scaling.

```

windows_spla = round( spla_rate )   if /windows/i matches vm.os
windows_spla = 0                     otherwise

units:   $/VM/month (flat, independent of RAM, disk and vCPU)
matches: "Microsoft Windows Server 2022", "windows server 2019", "WINDOWS 10"
does not: "Ubuntu Linux 22.04", "RHEL 9", blank / missing OS

```

2.5 Zerto disaster recovery — DR storage + replication fee

DR is opt-in **per VM**. It contributes two amounts, both zero unless that VM's Zerto toggle is on. The DR footprint `DR_Storage_GB` (journal + replica) is entered or imported manually and is **never derived from provisioned disk**.

```

if vm.dr is true:
    dr_storage = round( DR_Storage_GB × dr_storage_rate )
    dr_fee      = round( zerto_replication_fee )
else:
    dr_storage = 0 ; dr_fee = 0

zerto_dr = r2( dr_storage + dr_fee )    <- always rounded to cents

units:   GB × ($/GB/month) + $/protected VM/month
applies: only VMs whose Zerto toggle is on

```

Edge cases: (a) the DR subtotal is rounded to cents with `r2()` regardless of the rounding-mode setting; (b) a VM flagged as protected with 0 DR GB is still charged the flat replication fee; (c) toggling DR off leaves the stored DR GB value in place but contributes \$0.00 to both lines; (d) the DR footprint does not pass through the GB→TB divisor.

2.6 Add-ons

Add-ons are optional catalogue entries selected per VM. Each add-on has one of three unit types, which determines its quantity:

Unit type	Quantity	Formula	Divisor used?
per-vm	1	$\text{round}(1 \times \text{addon_rate})$	n/a
per-gb-ram	RAM_GB	$\text{round}(\text{RAM_GB} \times \text{addon_rate})$	No
per-tb-disk	$\text{Disk_GB} \div \text{divisor}$	$\text{round}((\text{Disk_GB} \div \text{divisor}) \times \text{addon_rate})$	Yes — same tbOf()

```
add_ons = round(  $\sum \text{round}( \text{qty}(a) \times \text{rate}(a) )$  for each selected add-on a )
```

Note: per-TB-disk add-ons always use the TB basis ($\text{Disk_GB} \div \text{divisor}$) — they ignore whether the VM's storage tier is priced per GB. Each add-on amount is rounded individually and the subtotal is rounded again. A selected add-on that has since been deleted from the catalogue is silently skipped.

3. Unit conversions

The model stores RAM and disk in GB. Only one conversion exists in the cost engine:

GB TO TB — `tbOf()`, `app.js`

```
tbOf(Disk_GB) = Disk_GB ÷ divisor          divisor ∈ {1024, 1000}, default 1024
```

```
1024  binary basis (TiB) — default
```

```
1000  decimal basis (TB)
```

The divisor affects: per-TB storage tiers, per-TB-disk add-ons, and the “total disk in TB” KPI. It does **not** affect: compute, VMware licensing, per-GB storage tiers, SPLA, or Zerto DR storage. If the setting is ever missing or non-numeric, the code falls back to 1024.

CSV import unit scaling

At import time the mapping modal lets each numeric column declare its source unit. Values are scaled to GB and rounded to two decimals before being stored, so all downstream math always operates on GB:

Column	Selectable units	Scale factor applied	Stored as
Disk	MB / GB / TB	$\text{MB} \times 1/1024 \cdot \text{GB} \times 1 \cdot \text{TB} \times 1024$	$\text{r2}(\text{value} \times \text{scale})$ GB
RAM	MB / GB	$\text{MB} \times 1/1024 \cdot \text{GB} \times 1$	$\text{r2}(\text{value} \times \text{scale})$ GB
DR storage	MB / GB / TB	$\text{MB} \times 1/1024 \cdot \text{GB} \times 1 \cdot \text{TB} \times 1024$	$\text{r2}(\text{value} \times \text{scale})$ GB

Import scaling always uses 1024 for MB/TB conversion — it is independent of the GB→TB billing divisor. Non-numeric cells are stripped of non-digit characters; if still not numeric they become 0 and a row-level warning is raised.

4. Rounding and currency display

A single helper performs all monetary rounding — half-up to two decimals with an epsilon nudge to counter binary floating-point representation error:

ROUNDING PRIMITIVE — `r2()`, `app.js`

```
r2(n) = Math.round( (n + Number.EPSILON) × 100 ) ÷ 100
```

Rounding is **not** display-only — it changes the stored numbers that the totals are built from. The rounding-mode setting selects which function wraps each line item:

Mode	Line items	Per-VM total	Effect
Round each line item (<code>rounding = 'line'</code> , default)	<code>round()</code> = <code>r2()</code> — each of compute, VMware, storage, SPLA, each add-on amount and the add-on subtotal is rounded to cents	<code>r2(sum of rounded lines)</code>	Totals are the exact sum of the cent-accurate line items shown in the table
Round totals only (<code>rounding = 'total'</code>)	<code>round()</code> = identity — line items keep full floating-point precision internally	<code>r2(sum of unrounded lines)</code>	A per-VM total may differ by up to a cent from adding up the displayed line items

Two amounts are always rounded with `r2()` irrespective of the mode: the Zerto DR subtotal (`dr = r2(dr_storage + dr_fee)`) and the per-VM grand total (`total = r2(...)`).

Display formatting is a separate, purely cosmetic layer: `usd()` renders values with `toLocaleString('en-US')` as USD currency with exactly two fraction digits; non-finite values render as \$0.00. Rate inputs are shown with more precision where needed — per-GB storage and DR rates display up to four decimals so a rate such as \$0.325/GB is not visually rounded away — but the underlying stored rate is used unmodified in the arithmetic.

5. Aggregations

All aggregates are plain sums of the per-VM values produced by `costVm()` — no additional rounding, weighting, tiering, minimum commit or volume discount is applied at any aggregation level.

Aggregate	Definition	Notes
Grand totals / KPI cards	Sum of each component across the currently visible rows; monthly cost = $\Sigma \text{total}(\text{vm})$	Respects the active location filter. “Annual” KPI = monthly $\times$ 12. Avg cost / VM = $\Sigma \text{total} \div \text{row count}$.
Per-location roll-up	VMs grouped by trimmed <code>vm.location</code> ; each component summed within the group	Blank or missing location groups under “Unassigned”. Share % = group total $\div$ grand total $\times$ 100, shown to one decimal.
SKU roll-up	One row per catalogue item actually in use: quantity = Σ of the driving quantity, amount = Σ of the matching per-VM line	Compute rows per ratio tier (Σ RAM GB); storage rows per tier (Σ TB or Σ GB per that tier's unit); SPLA counts Windows VMs; Zerto contributes two rows (Σ DR GB and protected-VM count); unused items are omitted.

Because each per-VM total is already rounded to cents in the default mode, a column total and the sum of its displayed cells always agree exactly.

6. Worked examples

Using the shipped sample inventory (sample-vm-inventory.csv) with the default placeholder rates, divisor = 1024, rounding = round each line item, VMware licensing enabled, and no add-ons configured. Every intermediate value below is reproducible by hand.

6.1 Inputs

VM	OS	Location	RAM GB	Disk GB	Ratio	Storage tier	Zerto	DR GB
WEB01	Microsoft Windows Server 2022	Columbus - DUB	16	200	4:1	Standard Flash	on	250
SQL01	Microsoft Windows Server 2019	Columbus - DUB	64	1024	2:1	High Performance Flash	on	1200
APP01	Ubuntu Linux 22.04	Indianapolis - 701 Congressional	32	500	4:1	Standard Flash	off	0

6.2 Step-by-step derivation

WEB01 — Windows, 4:1, per-TB Standard Flash, Zerto protected

```

compute      = r2( 16 × $10.00 )           = $160.00
vmware       = r2( 16 × $10.00 )           = $160.00
storage_qty  = 200 ÷ 1024                  = 0.1953125 TB
storage      = r2( 0.1953125 × $333.00 )   = r2($65.0390625) = $65.04
windows_spla = r2( $99.00 ) [ /windows/i matches OS] = $99.00
dr_storage   = r2( 250 × $0.15 )           = $37.50
dr_fee       = r2( $25.00 )                = $25.00
zerto_dr     = r2( 37.50 + 25.00 )          = $62.50
-----
total        = r2( 160.00+160.00+65.04+99.00+0.00+62.50 ) = $546.54

```

SQL01 — Windows, 2:1, per-TB High Performance Flash, Zerto protected

```

compute      = r2( 64 × $15.00 )           = $960.00
vmware       = r2( 64 × $10.00 )           = $640.00
storage_qty  = 1024 ÷ 1024                  = 1.0 TB
storage      = r2( 1.0 × $359.00 )          = $359.00
windows_spla = r2( $99.00 )                = $99.00
dr_storage   = r2( 1200 × $0.15 )           = $180.00
dr_fee       = r2( $25.00 )                = $25.00
zerto_dr     = r2( 180.00 + 25.00 )          = $205.00
-----
total        = r2( 960.00+640.00+359.00+99.00+0.00+205.00 ) = $2,263.00

```

APP01 — Linux, 4:1, per-TB Standard Flash, DR off

```

compute      = r2( 32 × $10.00 )           = $320.00
vmware       = r2( 32 × $10.00 )           = $320.00
storage_qty  = 500 ÷ 1024                  = 0.48828125 TB
storage      = r2( 0.48828125 × $333.00 )   = r2($162.59765625) = $162.60
windows_spla = 0 [no 'windows' in 'Ubuntu Linux 22.04'] = $0.00
zerto_dr     = 0 [Zerto toggle off]         = $0.00
-----
total        = r2( 320.00+320.00+162.60+0.00+0.00+0.00 ) = $802.60

```

6.3 Per-VM results and totals

VM	Compute	VMware lic.	Storage	SPLA	Add-ons	Zerto DR	Monthly total
WEB01	\$160.00	\$160.00	\$65.04	\$99.00	\$0.00	\$62.50	\$546.54
SQL01	\$960.00	\$640.00	\$359.00	\$99.00	\$0.00	\$205.00	\$2,263.00
APP01	\$320.00	\$320.00	\$162.60	\$0.00	\$0.00	\$0.00	\$802.60
Total (3 VMs)	\$1,440.00	\$1,120.00	\$586.64	\$198.00	\$0.00	\$267.50	\$3,612.14

Per-location roll-up for the same three VMs — sums of the per-VM totals above:

Location	VMs	RAM GB	Disk GB	Monthly total	Share
Columbus - DUB	2	80	1,224	\$2,809.54	77.8%
Indianapolis - 701 Congressional	1	32	500	\$802.60	22.2%
All locations	3	112	1,724	\$3,612.14	100%

The fourth sample row (FILE01 — Windows, 8 GB RAM, 1500 GB Standard Flash, DR off) adds \$80.00 compute + \$80.00 VMware + \$487.79 storage + \$99.00 SPLA = \$746.79, bringing the full four-VM sample inventory to \$4,358.93 per month.

7. Assumptions and limitations

- **Provisioned, not consumed.** Storage is billed on allocated disk as entered (Disk_GB). The model has no notion of thin-provision savings, deduplication, compression or actual utilisation.
- **Monthly recurring rates only.** Every rate is interpreted as a full-month charge. There is no proration for mid-month provisioning or decommissioning, no hourly or daily metering, and no partial-month credits. Annualised figures are simply the monthly total multiplied by 12.
- **Placeholder pricing.** The default catalogue values are illustrative. They must be replaced with the client's contracted SKU rates — including any negotiated discounts — before the output is used commercially.
- **RAM-based compute.** Compute and VMware licensing are functions of configured RAM only. vCPU count, CPU reservation, host overcommit and the ratio label itself do not enter the arithmetic; the ratio tier merely names the applicable per-GB rate.
- **SPLA by OS string.** Windows licensing is triggered solely by a case-insensitive “windows” substring in the OS field. A mistyped, abbreviated or blank OS value will silently omit the SPLA charge; a non-Windows VM whose OS text happens to contain “windows” will incur it.
- **DR footprint is manual.** DR GB is user-supplied and never inferred from provisioned disk, so its accuracy depends on the operator's Zerto journal and replica sizing.

- **No taxes, fees, or one-time charges.** The model excludes taxes, regulatory fees, bandwidth, backup, egress, professional services, hardware and any non-recurring setup costs. All amounts are USD; no FX conversion is applied.
- **Client-side and unversioned.** All inputs and rate cards live in browser localStorage. Figures reflect whatever the operator most recently entered; the app keeps no audit trail of rate changes, so quotes should be exported to CSV at the moment of issue.

8. Reproducing and auditing the figures

The CSV export is the audit artefact. It writes, for every VM, the inputs and every intermediate quantity used above — including `Disk_TB`, `StorageUnit`, `StorageBilledQty` and `StorageRate_perUnit` — one column per charge line, a TOTAL row, the per-location roll-up, and the pricing settings in force (including the GB→TB divisor). Any figure in the application can therefore be re-derived from the exported file using only the formulas in sections 2 to 4.

Audit step	What to check
1. Confirm the rate card	Compare every rate and SKU in the pricing tab against the contract. Defaults are placeholders (\$1).
2. Confirm the settings	GB→TB divisor (1024 or 1000) and rounding mode (per line item or totals only) — both change results.
3. Recompute a sample of VMs	Apply \$2 line by line; each line rounds to cents before the per-VM total is summed and rounded again.
4. Check conditional charges	SPLA appears exactly for VMs whose OS text contains “windows”; DR appears exactly for VMs with the Zerto toggle on, and the flat fee applies even at 0 DR GB.
5. Check the aggregates	Grand totals, location roll-up and SKU roll-up must equal the plain sums of the per-VM columns (\$5).

References

1. VM Cost Calculator (live application) — <https://vmcost-vmw.pplx.app>
2. Source repository, commit 8c75bee (app.js, README.md, sample-vm-inventory.csv) — <https://github.com/ryanwetmore492/vm-cost-calculator>
3. PapaParse 5.4.1 (CSV parsing, only third-party dependency) — <https://www.papaparse.com/>